

Module 07: Communication in Robotics — Bio-Inspired Multi-Robot Interaction

Learning Objectives

1. Explain how **biological communication systems** (pheromone trails, bee dances, bird flocking signals) inspire multi-robot communication architectures.
2. Analyze multi-robot communication as **coupled Active Inference** — where one robot's actions become another robot's observations.
3. Apply the concept of **generalized synchrony** to measure and optimize multi-robot coordination.

Introduction

No biological organism operates in complete isolation. Ants lay pheromone trails that guide colony-level foraging decisions. Bees perform waggle dances that communicate the distance and direction of food sources. Birds maintain flock cohesion through local visual and auditory signals. In each case, individual agents communicate by generating signals that modify other agents' sensory observations — exactly the multi-agent Active Inference architecture.

This module explores how bio-inspired communication gives multi-robot systems the capacity for coordination, cooperation, and emergent collective behavior.

Key Concepts

1. Stigmergic Communication: Pheromone Trails

Stigmergy is communication through environmental modification — leaving signals in the shared world rather than sending direct messages. Ant pheromone trails are the paradigmatic example:

- A robot that discovers a resource deposits a virtual "pheromone" (a spatial annotation in a shared map or a physical marker)

- Other robots detect the pheromone and update their generative models — the presence of pheromone increases the posterior probability that a resource exists in that direction
- Pheromone evaporation implements **natural forgetting** — stale information decays, preventing the system from converging on exhausted resources

This architecture is robust, decentralized, and scalable — no central planner is needed. Each robot performs local Active Inference with pheromone observations as additional sensory states.

2. Signal Dances: Direct Information Transfer

The honeybee waggle dance encodes symbolic information (direction and distance to food) in movement. Robotic analogs implement **direct information transfer**:

- A scouting robot encodes the location and quality of a discovered resource into a compact message (coordinates, confidence level, resource type)
- Receiving robots update their generative models with this message, weighted by **trust precision** — a reliability estimate of the sender
- The receiving robot's posterior is a precision-weighted combination of its prior beliefs and the message content

This is formally identical to **Bayesian belief updating with social information** — the same mathematics that describes human testimony and expertise.

3. Flocking as Coupled Inference

Bird flocking (Reynolds' rules: separation, alignment, cohesion) is naturally formulated as coupled Active Inference:

- Each bird/robot maintains a generative model that predicts the positions and velocities of its neighbors
- **Separation**: Prediction errors from neighbors being too close drive avoidance actions
- **Alignment**: Prediction errors from velocity mismatches drive heading corrections
- **Cohesion**: Prediction errors from neighbors being too far drive approach actions

- The precision on each prediction error determines the relative weighting of separation, alignment, and cohesion — adjustable parameters that produce different flock configurations

Emergent flock behavior arises from local inference — no robot explicitly represents the "flock shape." The global pattern emerges from coupled local free energy minimization.

4. Heterogeneous Communication

Real multi-robot systems often include **heterogeneous agents** — ground robots communicating with aerial drones, or slow sensor nodes communicating with fast actuator robots. Bio-inspired heterogeneous communication:

- Different agent types contribute different observation modalities to the shared inference (aerial robots provide maps; ground robots provide detailed object information)
- Information quality varies by agent type (precision weighting handles this automatically)
- Mixed-bandwidth communication channels (high-bandwidth local, low-bandwidth remote) are modeled as observations with different precision

Applications

- **Search and rescue swarm:** A multi-robot team searches a disaster site using stigmergic communication — robots that locate survivors deposit high-confidence markers, attracting nearby robots to converge on the location. Pheromone evaporation ensures the swarm doesn't persist at already-rescued locations.
- **Multi-drone crop monitoring:** Aerial drones communicate field condition observations to ground-based treatment robots using signal-dance protocols, with trust precision calibrated by sensor quality. The fleet collectively builds a generative model of crop health across the entire field.

Conclusion

Bio-inspired communication — stigmergy, signal transfer, flocking, and heterogeneous coordination — enables multi-robot systems to achieve collective intelligence through local

Active Inference. The next module examines how bio-inspired planning gives robots the capacity for deliberative, long-horizon behavior.